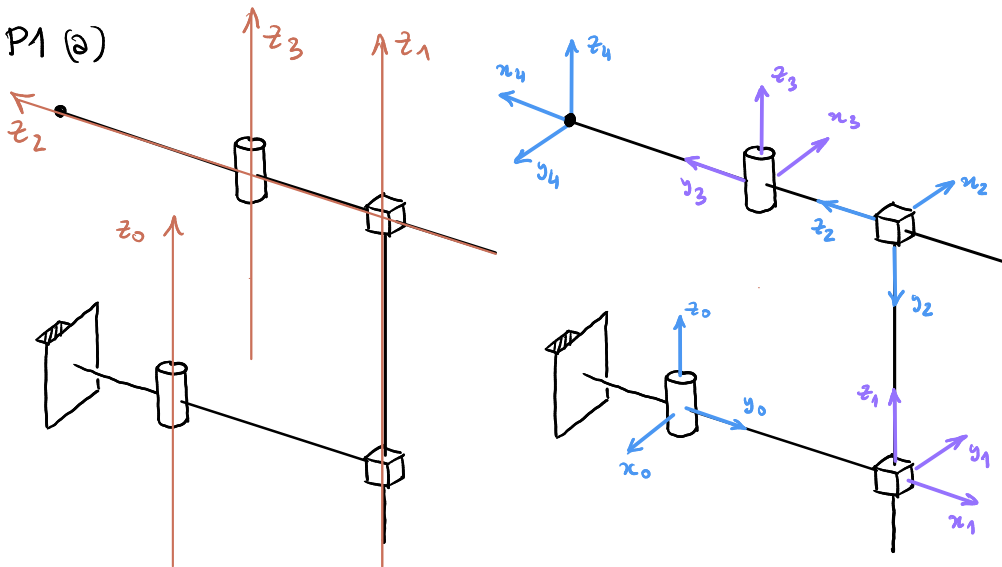
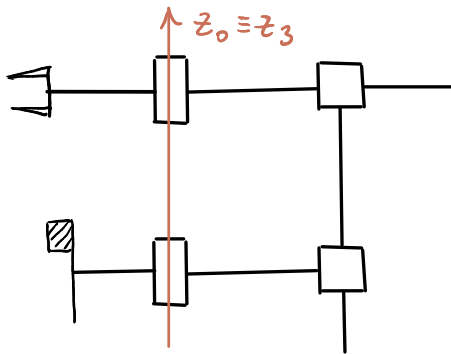


P1 (a)



d_i	θ_i	a_i	α_i
0	$\theta_1(t)$	a_1	0
$d_2(t)$	$\pi/2$	0	$-\pi/2$
$d_3(t)$	0	0	$\pi/2$
0	$\theta_4(t)$	a_4	0

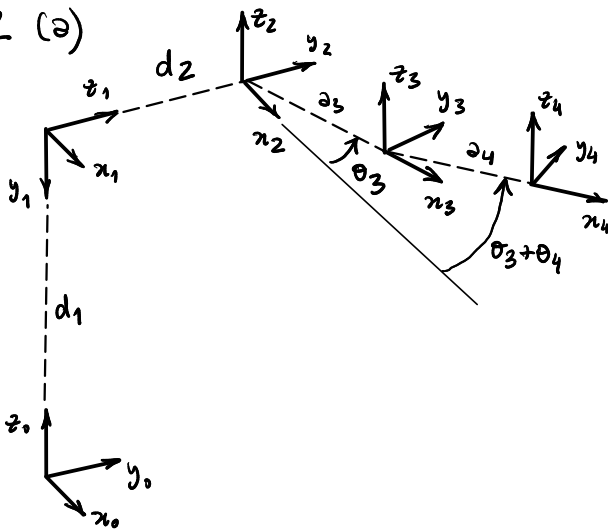
(b)



Internal singularity :

$$z_0 \equiv z_3, \quad d_3 = a_1.$$

P2 (a)



$$R_4^0 = R_1^0 R_2^1 R_3^2 R_4^3 = \cancel{R_2^2}^I R_4^2$$

$$= \begin{pmatrix} c_{34} & -s_{34} & 0 \\ s_{34} & c_{34} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$p_4 = d_1 z_0 + d_2 y_0 + a_3 x_3 + a_4 x_4$$

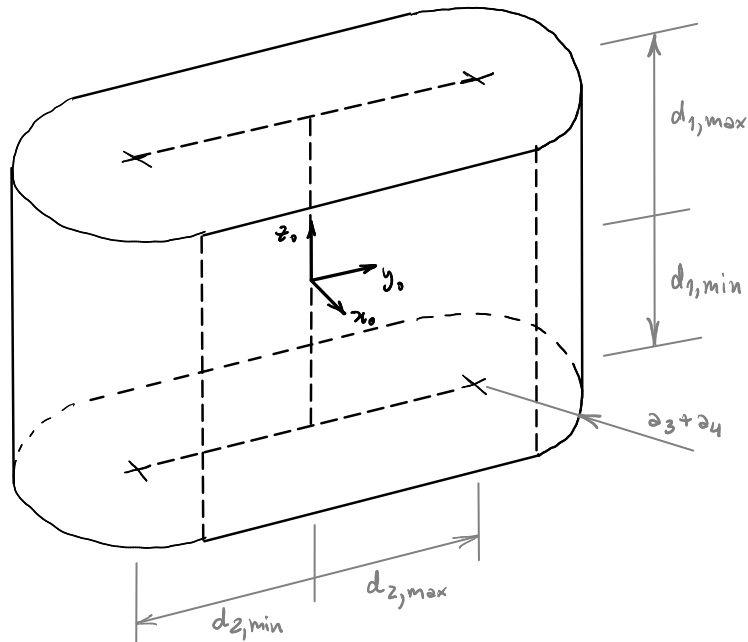
$$= \begin{pmatrix} a_3 c_3 + a_4 c_{34} \\ d_2 + a_3 s_3 + a_4 s_{34} \\ d_1 \end{pmatrix}.$$

$$(b) \quad \mathcal{Y}_p = \frac{\partial p_4}{\partial q} = \begin{pmatrix} 0 & 0 & -a_3 s_3 - a_4 s_{34} & -a_4 s_{34} \\ 0 & 1 & a_3 c_3 + a_4 c_{34} & a_4 c_{34} \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

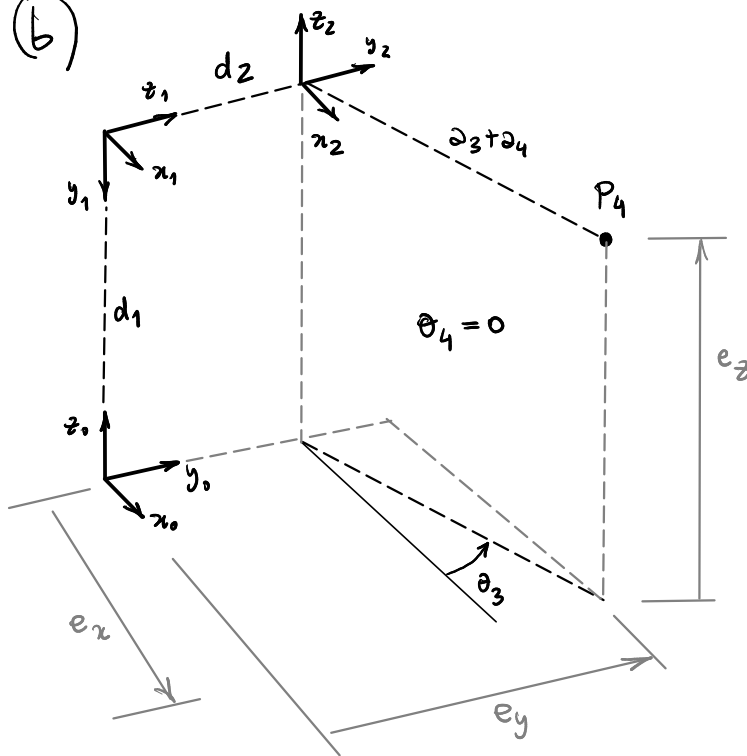
$$\mathcal{Y}_0 = \begin{pmatrix} 0 & 0 & z_2 & z_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

$$\mathcal{Y} = \begin{pmatrix} \mathcal{Y}_p \\ \mathcal{Y}_0 \end{pmatrix}.$$

P3 (a)



(b)



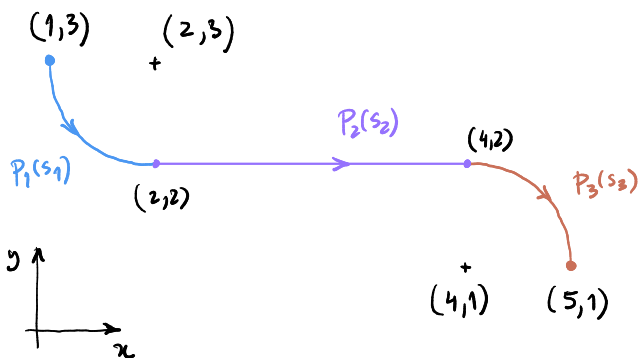
$$\text{Let } (P_4)_d \triangleq (e_x \ e_y \ e_z)^T$$

$$d_1 = e_z.$$

$$(\partial_3 + \partial_4) \cdot e_z = e_x, \\ \Rightarrow \theta_3 = \arccos\left(\frac{e_x}{\partial_3 + \partial_4}\right).$$

$$d_2 = e_y - (\partial_3 + \partial_4) s_3.$$

P4 (a)



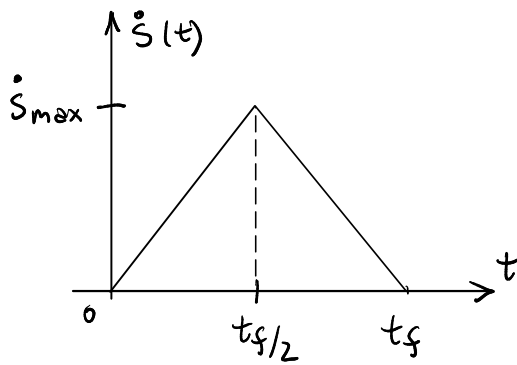
$$P_1(s_1) = \begin{pmatrix} 2 - \cos(s_1) \\ 3 - \sin(s_1) \end{pmatrix}, \quad 0 \leq s_1 \leq \pi/2$$

$$P_2(s_2) = \begin{pmatrix} 2 + s_2 \\ 2 \end{pmatrix}, \quad 0 \leq s_2 \leq 2$$

$$P_3(s_3) = \begin{pmatrix} 4 + \sin(s_3) \\ 1 + \cos(s_3) \end{pmatrix}, \quad 0 \leq s_3 \leq \pi/2$$

$$P(s) = \begin{cases} P_1(s) & , \quad 0 \leq s < \pi/2 \\ P_2(s - \pi/2) & , \quad \pi/2 \leq s < \pi/2 + 2 \\ P_3(s - \pi/2 - 2) & , \quad \pi/2 + 2 \leq s \leq \pi + 2 \end{cases}$$

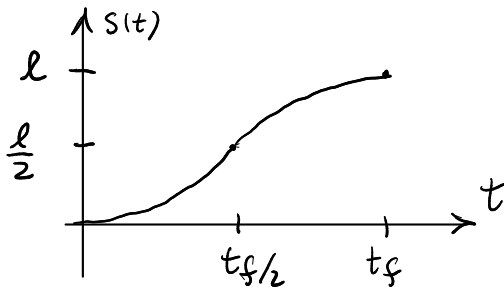
P4 (b)



$$l = \int_0^{t_f} \dot{s}(u) du \stackrel{\text{Area}}{=} \frac{t_f \dot{s}_{max}}{2}.$$

$$s(t) = \begin{cases} at^2 & , 0 \leq t < t_f/2 \\ l - a(t - t_f)^2 & , t_f/2 \leq t \leq t_f \end{cases}$$

$$\frac{l}{2} = a \left(\frac{t_f}{2} \right)^2 \Rightarrow a = \frac{2l}{t_f^2} = \frac{\dot{s}_{max}}{t_f}$$



$$s(t) = \begin{cases} \frac{\dot{s}_{max}}{t_f} t^2 & , 0 \leq t < t_f/2 \\ \frac{t_f \dot{s}_{max}}{2} - \frac{\dot{s}_{max}}{t_f} (t - t_f)^2 & , t_f/2 \leq t \leq t_f \end{cases}$$

(c) $l = \frac{t_f \dot{s}_{max}}{2} \Rightarrow 2(\pi + 2) = t_f \dot{s}_{max}.$